Supriya Veerla

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AI Engineer | Generative AI Developer

EXPERIENCE

Teaching Assistant **January 2026 - Present**
Department of Data Science, University of North Texas

- Serving as a Teaching Assistant for graduate-level data science courses under Dr. Mohammed Aledhari, supporting instruction in machine learning and generative AI.
- Assist students with model development, evaluation, and practical implementation while reinforcing responsible AI practices.

Research Assistant **May 2025 - Present**
Department of Data Science, University of North Texas

- Working on AI-driven healthcare projects using wearable sensor data such as glucose and physiological signals.
- Developed and evaluated machine learning models to predict health risks and support timely interventions.

AI-ML Virtual Intern **January 2024 - March 2024**
Google for Developers

- Streamlined data workflows by architecting a data ingestion pipeline using Pandas and NumPy, slashing manual data processing errors by 20% and saving the team 3 analyst hours per week.
- Pinpointed five previously unidentified drivers of customer churn through rigorous analysis of user behavior data, leading to a 4% improvement in overall customer retention rates.
- Piloted the integration of LLM workflows with FAISS/Pinecone for vector retrieval in prototype pipelines, contributing to RAG-based system experimentation, and improving contextual accuracy by 27% during internship.

Data Analyst Intern **July 2023 - December 2023**
CS CODENZ

- Processed and analyzed large-scale educational datasets using Python, SQL, and SAP, improving data integrity and preprocessing efficiency by 25%.
- Visualized AP EAPCET rank cut-off trends with Tableau and Matplotlib, delivering interactive dashboards to 63+ stakeholders and enabling quick data-driven decision-making for resource allocation.
- Led comprehensive data cleaning, transformation, and feature engineering pipelines, boosting predictive accuracy by 12% across experiments.
- Devised a series of reusable data pipelines and preprocessing scripts that are aligned with the team's MLOps standards, cutting down deployment time by 31% and enabling scalable experimentation.
- Applied exploratory data analysis and ML model evaluation metrics to validate dataset quality, resulting in 17% improvement in model accuracy.
- Integrated digital marketing campaign data with sales performance metrics, providing insights that improved lead conversion rates by 40%.

PROJECTS

Gen AI-Powered LinkedIn Post Generator using Llama 3.2 **September 2025 – October 2025**

- Forged a GenAI-powered LinkedIn post generator utilizing Llama 3.2, LangChain, and Groq Cloud, maintaining style consistency above 90% and adhering to brand guidelines.
- Engineered a data ingestion and preprocessing pipeline using Python, Pandas, and NumPy, implementing prompt engineering to achieve 39% improvement in tone accuracy.
- Masterminded interactive Streamlit interface with Groq LPU inference, trimming generation turnaround by 70% against CPU inference and mentoring 2 junior engineers.
- Constructed custom prompt templates and output validators using LangChain, improving output relevance and reducing hallucination by 29%.
- Optimized generative pipelines through content structuring and parameter tuning, improving coherence and fluency, reflected in an 65% improvement in perplexity score.
- Delivered a production-ready AI tool for generating LinkedIn posts, reducing manual drafting time by 50-80% with automated content generation.
- Transitioned manual data processes to a fully automatic Power BI dashboard with scheduled refreshes, curtailing report generation time by 12 hours a week and improving data availability.

Tools/Tech : Llama 3.2, LangChain, Groq Cloud, Streamlit, Python, Hugging Face Transformers, Prompt Engineering, Few-Shot Learning, Vector Embeddings, Pandas, NumPy, Git, GitHub, Cloud Deployment, Generative AI, Model Optimization

AI-Powered Healthcare Chatbot using LLM and RAG **March 2025 – April 2025**

- Orchestrated the healthcare chatbot using OpenAI GPT models and a full RAG architecture, enabling context-aware medical query responses; findings led to identifying and fixing the three biggest causes of crashes.
- Revamped the backend infrastructure with AWS Lambda and API Gateway, which accelerated real-time conversational capabilities while slashing server costs by $25,000 annually and enabling 99.9% API request success.
- Fortified a Streamlit front-end with RAG pipelines and encrypted API calls, ensuring HIPAA-aligned handling of sensitive healthcare data; slashed potential data breaches by 57%.
- Implemented advanced prompt tuning, contextual grounding, and hallucination-reduction techniques, improving response reliability and medical relevance for patient-facing queries by 43%.
- Revolutionized the GenAI system's modular pipelines, implementing vector indexing to optimize API orchestration and continuous performance evaluation, while maintaining a consistent throughput of 20+ queries per second.

Tools/Tech : OpenAI GPT API, FAISS, LangChain, RAG Architecture, Streamlit, AWS Lambda, API Gateway, Python, SQL, Pandas, Vector Databases, Semantic Search, Embeddings, REST APIs, JSON, Cloud Deployment, Secure Data Handling, Prompt Engineering, LLMOps

Hospital Emergency Room Dashboard **December 2024 – February 2025**

- Designed and deployed an interactive Power BI dashboard for real-time ER monitoring, integrating Azure/SharePoint for secure access, resulting in 39% increase in user adoption.
- Automated data validation processes within ETL pipelines using Python, reducing manual effort by 7 hours weekly, and improving overall data quality by roughly 20%.
- Analyzed patient demographics, triage, and referral patterns to drive data-backed decisions, reducing bottlenecks and saving 29 hours per week.

Tools/Tech : Power BI, SQL, Power Query, DAX, Excel, Python, Pandas, Data Visualization, ETL Pipelines, Azure, SharePoint, Forecasting, KPI Analytics, Data Modeling, Reporting Automation

SKILLS

AI/ML Core Machine Learning, Deep Learning, Supervised Learning, Unsupervised Learning, Neural Networks, Model Training, Model Evaluation, Feature Engineering

Generative AI & LLMs Large Language Models (LLMs), LangChain, Prompt Engineering, RAG (Retrieval-Augmented Generation), Fine-tuning (LoRA, QLoRA), Embeddings, Tokenization, Vector Databases

Cloud & MLOps AWS (SageMaker, Lambda, Bedrock, S3), GCP (Vertex AI), Azure ML, Docker, Kubernetes, CI/CD, Model Deployment, ML Pipelines

Data Engineering & Visualization Pandas, NumPy, PySpark, Data Pipelines, Data Cleaning, Data Transformation, Exploratory Data Analysis (EDA), KPI Dashboards

EDUCATION

Master of Science in Data Science **August 2024 - May 2026**
University of North Texas Denton, Texas

Bachelor of Technology in Computer Science and Engineering **January 2021 - May 2024**
Sri Vasavi Engineering College, Tadepalligudem Andhra Pradesh, India